



TÉCNICO
LISBOA

Multimodal Data of the 2026 Portuguese Presidential Elections

Exploratory Data Analysis

Group 25:

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Processing Big Data

Data Overview



Visual

- Bounding boxes + confidence
- Poses
- Facial emotions (8 Classes)



Audio/Speech

- Acoustic Features (Pitch, jitter)
- Speaker diarization
- Text transcripts & embedding



Original source

- 28 political debates
- 7 newscasts
- Multi source polling data

Visual data noise

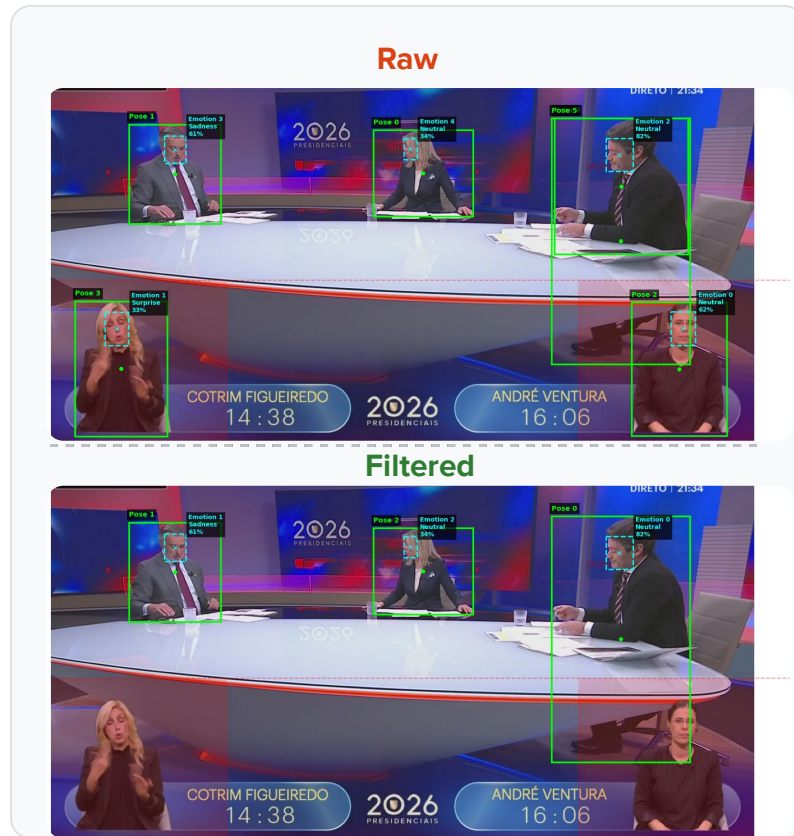
- Unsorted entries → Sorted entries by n° of frame

Noise Sources

- Sign language interpreters
- Bounding box overlapping duplicates

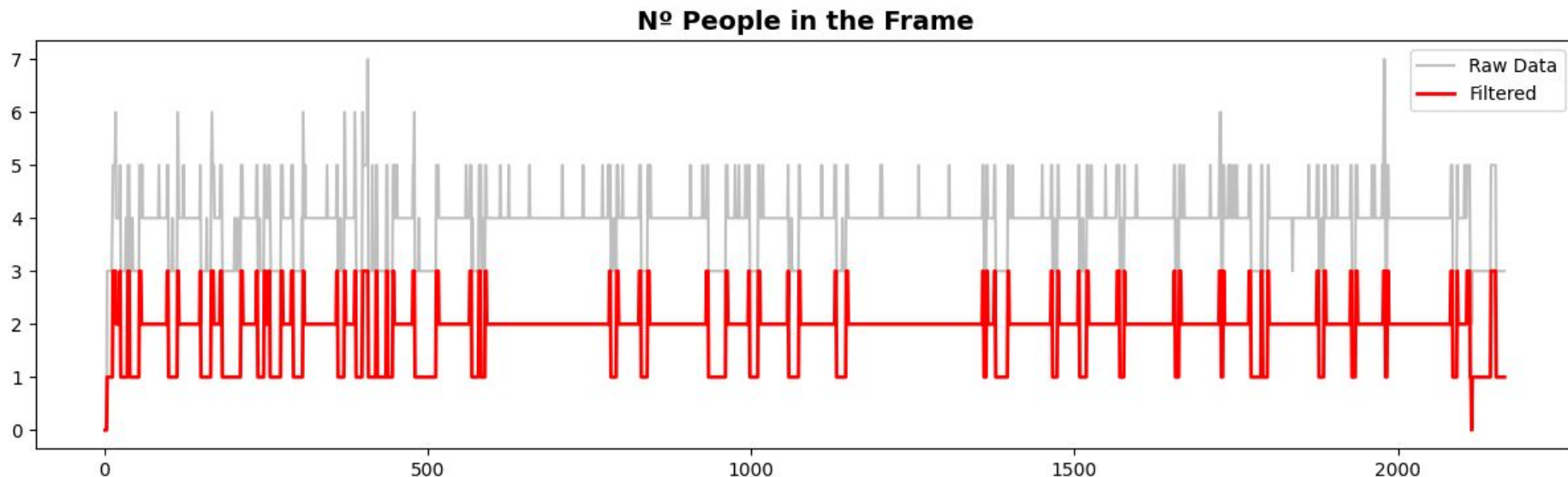
Solution

- Excluding corner dead zones
- Confidence threshold $> 40\%$
- Exclusion by overlapping
- Exclusion by area size of bounding box



Visual data noise

Example for Cotrim Figueiredo vs Ventura debate on December 19



- NºPeople = length of cleaned poses vector

Missing Data

Broadcast Realities

"0 faces" detected usually means commercials or camera moves.

Alignment Gaps

Desynchronization between audio transcripts and visual frames.

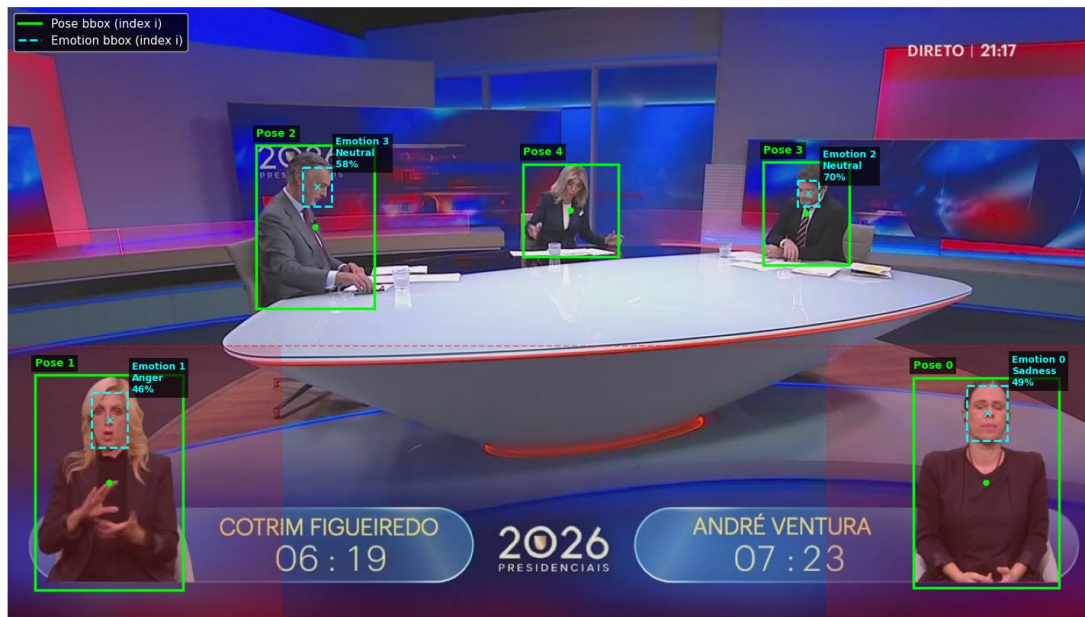
Models Failure

No emotions detected or no transcript of some sections

Possible Solution

Regression Fill?

Missing Data



2D Visualization of Audio Characteristics

Final Goal: How to identify a segment transcript speaker using the audio

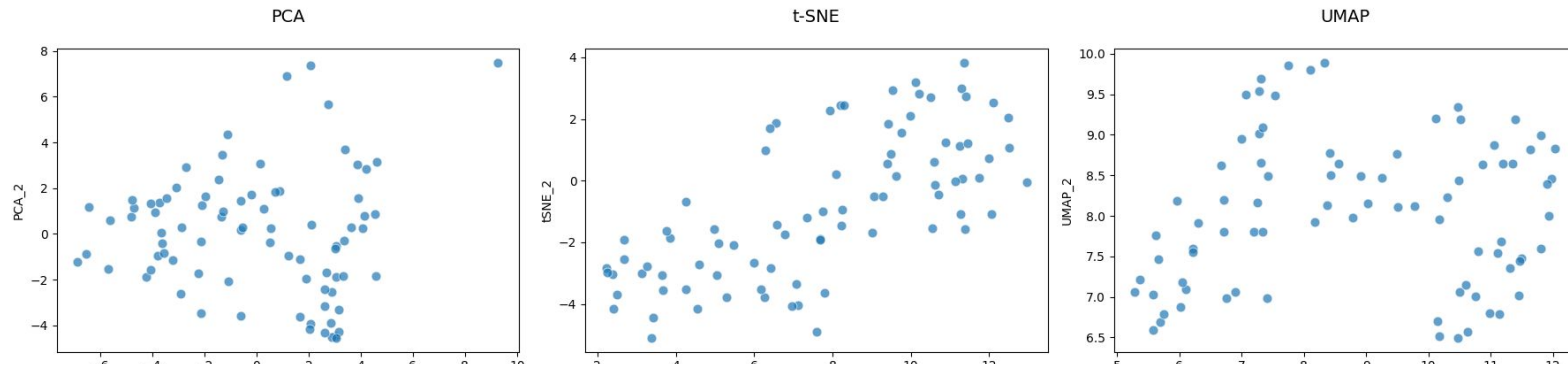
- Which audio characteristics are more ideal to make this?
 - Simple acoustic features: pitch, speak rate,...
 - Pyannote speaker embeddings

Use dimensionality reduction to 2D to look for possible Clusters

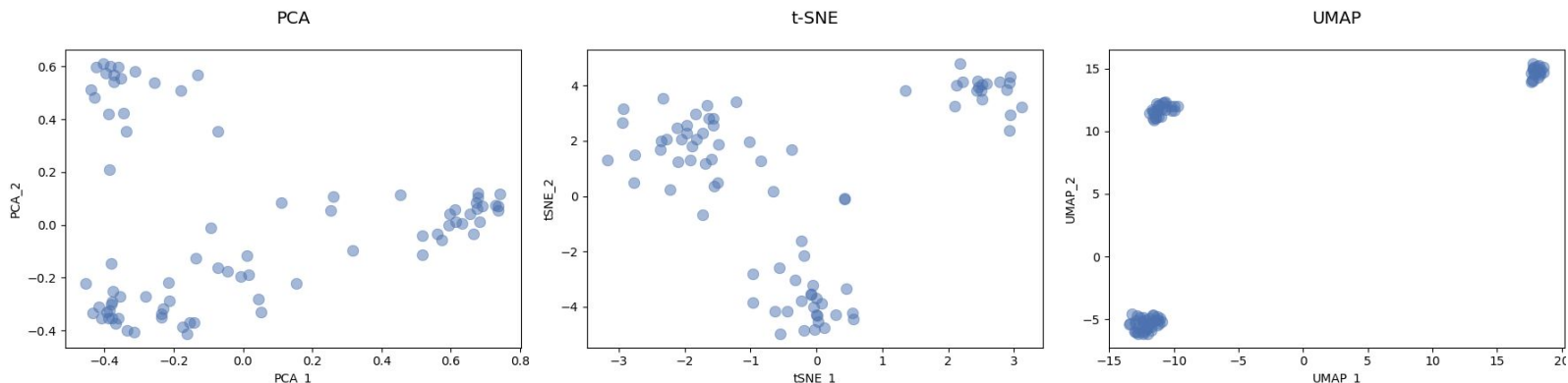
- Which algorithm to use?
 - Simple PCA
 - T-SNE
 - UMAP

2D Visualization of Audio Characteristics

Dimensionality Reduction: Only Acoustic Features (30D -> 2D)



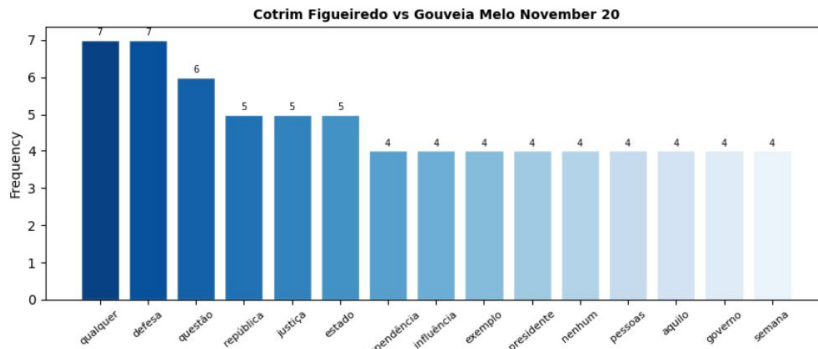
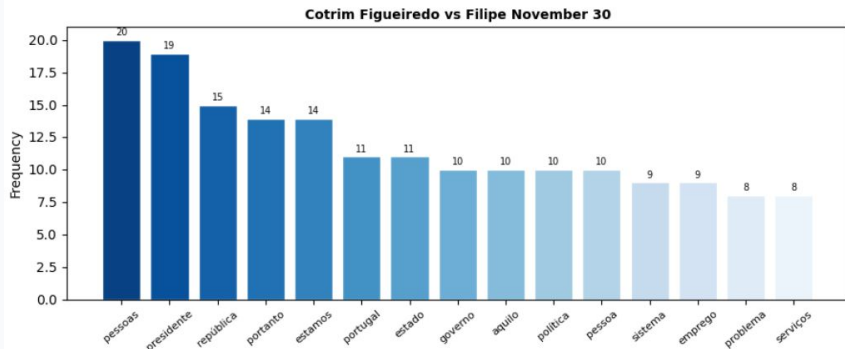
Dimensionality Reduction: Speaker Embeddings (512D -> 2D)



Transcript Analysis

Simple frequency histograms or box plots doesn't tell us much about what's going on

Most Common Words per Debate (Bigger than 5 chars)



What if we can tell the topic of a random segment of the transcript, but how?

Transcript Analysis

"a priori" themes common in presidential debates

Estabilidade Política

Habitação

Imigração

Economia

Defesa

Juventude

Justiça / Corrupção

Crise Ambiental

Eleições

Ex: SNS / Saúde

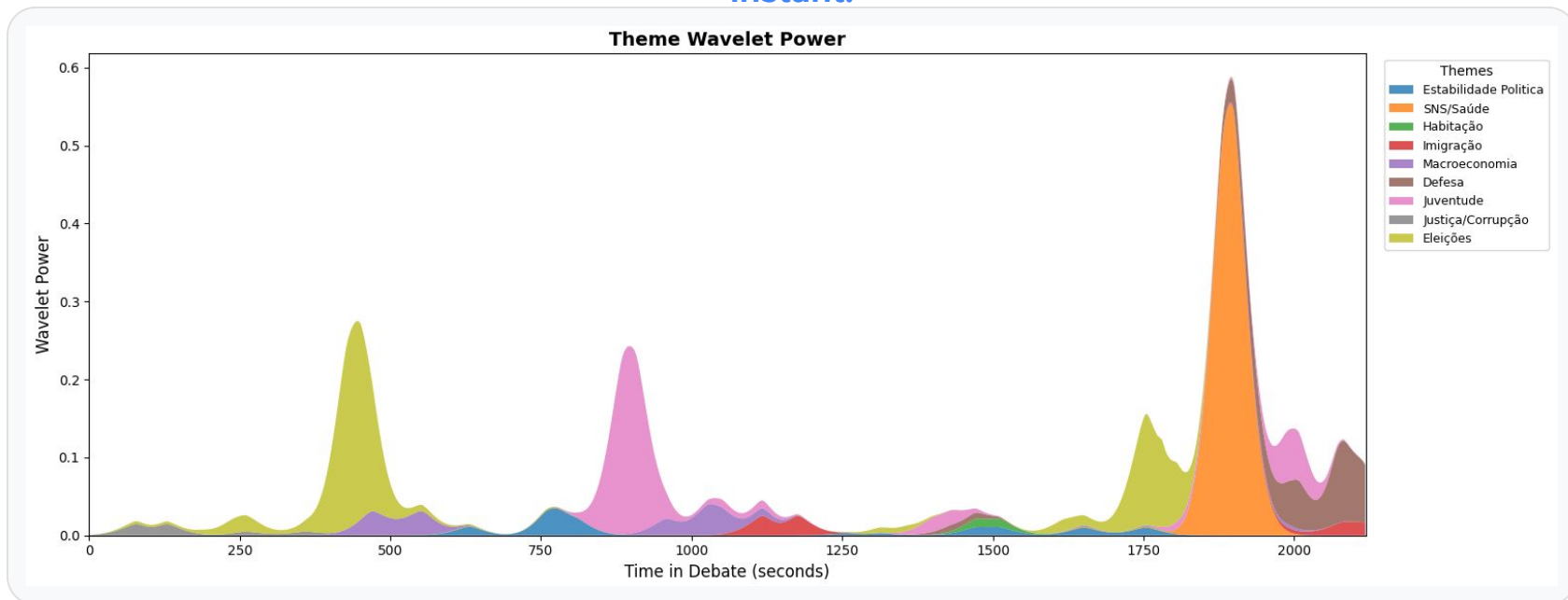
Catch words:

SNS, médicos, urgências, hospitalais, centros de saúde, enfermeiros, listas de espera

Transcript Analysis

But now we have another problem: we need a tool that lets us see frequency and time analysis at the same time, avoiding predetermined intervals.

To solve this, we can use **Wavelet Analysis** for each theme and see the leading one for each instant:





Thank you !